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14MY 44 – IV (35)

IV Semester B.Sc. Degree Examination, May 2014
COMPUTER SCIENCE (New)
BSC 404 CS : Data Structures Using 'C'

Time : 3 Hours Max. Marks : 80

Instruction : ***All*** *the questions are* ***compulsory****.*

SECTION – A

I. Answer **any ten** of the following : (10×2=20)

1) Why do you need to allocate memory at run-time ?

2) Where do you find calloc () function ?

3) What is a STACK ?

4) What is the significance of the top in a stack ?

5) How is the queue different from the stack ?

6) What is a deque ?

7) What are the basic components of a linked list ?

8) What is the benefit of using circular doubly linked list ?

9) What is a tree ?

10) Define :

   a) Preorder traversal.

   b) Postorder traversal.

11) What is the need for sorting ?

12) What is hasting ?

P.T.O.

SECTION – B

II. Answer **any six** of the following : (6×5=30)

13) Justify : Algorithm + Data structure = Program.

14) Write a C-Program for linear searching.

15) Explain the different operations to be performed on data structures.

16) Evaluate the following post fix expressions :

    AB + CD – ∗, given A = 1.0, B = 2.0, C = 3.0 and D = 4.0.

17) How is the queue different from the stack ?

18) What is a circular queue ? How do you represent it ?

19) List the different types of linked lists and give their structure definition.

20) Explain the three binary tree traversal methods.

SECTION – C

III. Answer **any three** of the following : (3×10=30)

21) What is Queue ? Explain the procedure to perform different operations on it.

22) Write a algorithm for linear search and binary search method.

23) Explain all the operations performed on binary tree.

24) Explain merge sort with an example.

25) Explain Deque. Write an algorithm to insert an ITEM at the FRONT end of Deque.